Hari Patel

267-905-5972 | Philadelphia, PA | haririteshpatel@gmail.com | linkedin.com/in/haripatel788 | haripatel788.github.io

Education

Drexel University | Pennoni Honors College — Philadelphia, PA
B.S. Computer Science; Minor in Artificial Intelligence & Machine Learning — *Expected Jun. 2030*
- **GPA:** 3.88/4.00 **Honors:** Presidential Scholarship; Dean's List (Fall, Winter, Spring 2025–26)
- **Coursework:** Computer Programming I & II, Software Engineering, Multivariate Calculus, Linear Algebra

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, FastAPI, Express.js, Tailwind CSS, Scikit-Learn, Pandas, NumPy
Tools & Databases: Git, GitHub, PostgreSQL, Supabase, Firebase, Vercel, AWS (EC2, RDS), pytest
Concepts: Data Structures & Algorithms, OOP, REST API Design, Database Design, Authentication & Authorization, Unit Testing, Machine Learning, LLM Integration
Practices: Full-Stack Development, Agile, Code Review

Experience

Undergraduate Researcher | *Drexel University (Advisor: Dr. Arvin Ebrahimkhanlou)* — Oct. 2026 – Present
- Benchmarked structural condition diagnosis models across CVPR literature, selecting candidate Generative AI architectures for multi-modal causal reasoning pipelines.

Co-Founder, Novus | *React, TypeScript, Supabase, PostgreSQL, Llama 3.3 70B* — Apr. 2026 – Present
- Architected a multi-tenant backend on Supabase utilizing a 40-table PostgreSQL schema with Row-Level Security (RLS) to enforce strict institutional data isolation and role-based access control.
- Built an automated grading pipeline using Llama 3.3 70B via Supabase Edge Functions to evaluate technical accuracy, risk awareness, and communication, achieving a 92.5% alignment rate with expert evaluation benchmarks within a 2-point delta.
- Partnered with 8+ finance industry advisors to author 80+ rigorous technical interview questions; engineered real-time scoring endpoints for beta deployment.

Software Engineering Mentee, CodeHS — Jun. 2025 – Sep. 2025
- Participated in code reviews, pull request workflows, and sprint ceremonies under the direct mentorship of an 8-person production engineering team.

Projects

Market Pulse | *Python, FastAPI, Pandas, Supabase* — Feb. 2026 – Present
- Engineered a stock volatility forecasting REST API using per-stock Heterogeneous Autoregressive (HAR) models across 10 years of price history to calculate expected daily moves and 80%/95% prediction intervals.
- Implemented Supabase JWT authentication, tiered RBAC, sliding-window rate limiting, and an automated test suite containing 27 pytest integration tests.
- Backtested 112,700 predictions across 490 S&P 500 equities, outperforming EWMA baselines on 86% of 474 held-out stocks across 1-, 5-, and 20-day horizons with an 81% empirical coverage rate on 80% confidence bands.

Sixth & Market | *React, TypeScript, Express.js, Firebase, Groq API, Leaflet* — Apr. 2026
- Awarded 9th place out of 52 teams at Philly Codefest 2026 for engineering and deploying a civic analytics application within an 11-hour hackathon sprint.
- Integrated Groq-accelerated LLM endpoints via an Express.js reverse proxy to classify Philadelphia 311 service tickets while securing API credentials server-side.
- Built an interactive geospatial Leaflet dashboard cross-referencing FCC broadband coverage, U.S. Census tracts, and municipal complaint datasets to highlight connectivity dead zones.

PrepGo | *Next.js, TypeScript, Tailwind CSS, PostgreSQL (Neon), Clerk, Vercel* — Jan. 2026 – Mar. 2026
- Deployed a full-stack SAT preparation platform serving 30 active users and 300+ completed practice questions with Clerk authentication.
- Engineered a per-session rate limiter to mitigate connection pool exhaustion and resolve high-concurrency server crashes under peak load.
- Developed the frontend interface in Next.js/Tailwind CSS and configured serverless PostgreSQL database queries on Neon.

Leadership & Volunteering

Kishore Coordinator — Sep. 2025 – Present
Bochasanwasi Shri Akshar Purushottam Sanstha — *Philadelphia, PA*
- Coordinated weekly programs for high-school and college youth, driving a 36% increase in average attendance over a 52-week period.
- Organized 7 large-scale community events, directing personal outreach initiatives to successfully integrate newcomers.